

CS331 SYSTEM SOFTWARE LAB**COET****DEPT OF CSE****1. INTRODUCTION TO OPERATING SYSTEM**

An operating system (OS) is the software component of a computer system that is responsible for the management and coordination of activities and the sharing of the resources of the computer. The OS acts as a host for application programs that are run on the machine. As a host, one of the purposes of an OS is to handle the details of the operation of the hardware. This relieves application programs from having to manage these details and makes it easier to write applications. Almost all computers use an OS of some type.

OSs offer a number of services to application programs and users. Applications access these services through application programming interfaces (APIs) or system calls. By using these interfaces, the application can request a service from the OS, pass parameters, and receive the results of the operation. Users may also interact with the OS by typing commands or using a graphical user interface (GUI).

2. SCHEDULING ALGORITHM

CPU Scheduling is a process of determining which process will own CPU for execution while another process is on hold. The main task of CPU scheduling is to make sure that whenever the CPU remains idle, the OS at least select one of the processes available in the ready queue for execution. The selection process will be carried out by the CPU scheduler. It selects one of the processes in memory that are ready for execution.

a) PROCESS SCHEDULING

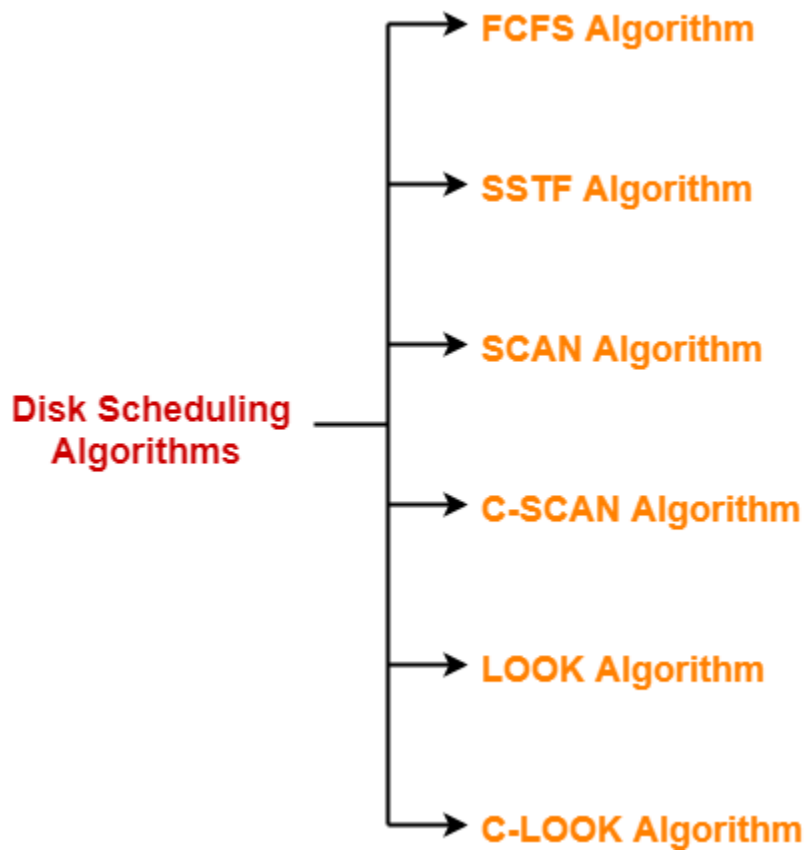


The process scheduling is the activity of the process manager that handles the removal of the running process from the CPU and the selection of another process on the basis of a particular strategy.

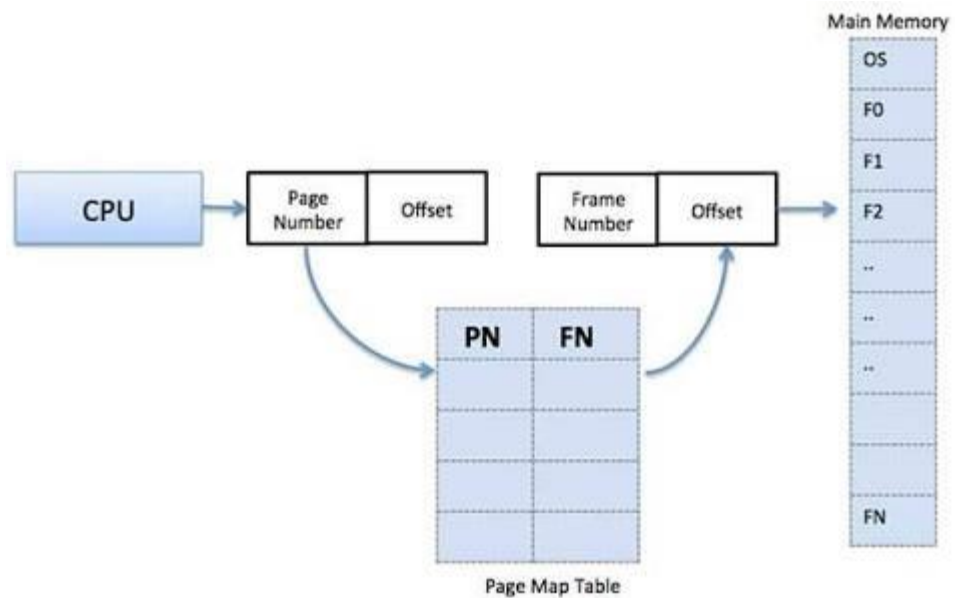
Process scheduling is an essential part of a Multiprogramming operating systems. Such operating systems allow more than one process to be loaded into the executable memory at a time and the loaded process shares the CPU using time multiplexing.

b) DISK SCHEDULING

- Disk scheduling is a technique used by the operating system to schedule multiple requests for accessing the disk. The algorithms used for disk scheduling are called as disk scheduling algorithms.
- The purpose of disk scheduling algorithms is to reduce the total seek time.
- Various disk scheduling algorithms are-



3. MEMORY MANAGEMENT



Memory management is the process of controlling and coordinating computer memory, assigning portions called blocks to various running programs to optimize overall system performance. Memory management resides in hardware, in the OS (operating system), and in programs and applications.

In hardware, memory management involves components that physically store data, such as RAM (random access memory) chips, memory caches, and flash-based SSDs (solid-state drives). In the OS, memory management involves the allocation (and constant reallocation) of specific memory blocks to individual programs as user demands change. At the application level, memory management ensures the availability of adequate memory for the objects and data structures of each running program at all times. Application memory management combines two related tasks, known as allocation and recycling.

4. **SEMAPHORE**

Semaphore is simply a variable that is non-negative and shared between threads. A semaphore is a signaling mechanism, and a thread that is waiting on a semaphore can be signaled by another thread. It uses two atomic operations, 1)wait, and 2) signal for the process synchronization.

A semaphore either allows or disallows access to the resource, which depends on how it is set up.

5. **DEADLOCK**

Deadlock is a situation where a set of processes are blocked because each process is holding a resource and waiting for another resource acquired by some other process.

Deadlock can arise if following four conditions hold simultaneously (Necessary Conditions)

- **Mutual Exclusion:** One or more than one resource are non-sharable (Only one process can use at a time)
- **Hold and Wait:** A process is holding at least one resource and waiting for resources.
- **No Preemption:** A resource cannot be taken from a process unless the process releases the resource.
- **Circular Wait:** A set of processes are waiting for each other in circular form.

6. **DEADLOCK AVOIDANCE**

- Deadlock prevention causes low device utilization and reduced system throughput.
- To avoid deadlocks, one requires additional information about how resources all to be requested.
- For one instance of each resource type, Resource Allocation Graph Algorithm is used.
- For avoidance of multiple instances of each resource type, Banker's Algorithm is used.
- The banker's algorithm is a resource allocation and deadlock avoidance algorithm that tests for safety by simulating the allocation for predetermined maximum possible amounts of all resources, then makes an "s-state" check to test for possible activities, before deciding whether allocation should be allowed to continue.